# From force diagrams to motion

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a force diagram to predict acceleration and velocity change.

**SUCCESS CRITERIA - I can...**

1. Write a force sum along each chosen axis.
2. Connect net force and mass to acceleration.
3. Use velocity direction to decide the initial speed change.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “parallel” and “perpendicular” to explain your answer. Sentence starters are optional.

**parallel**: along the same direction as a line or surface

**perpendicular**: at a right angle, 90°, to a line or surface

**component**: the part of a vector along one chosen axis

## 01 Warm-up: bring back an earlier idea

A **3 kg** cart has **5 N** right and **26 N** left.
1. **Find acceleration**.
2. **What else** must you know to decide if it speeds up?

**Word help:** acceleration = rate of change of velocity

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

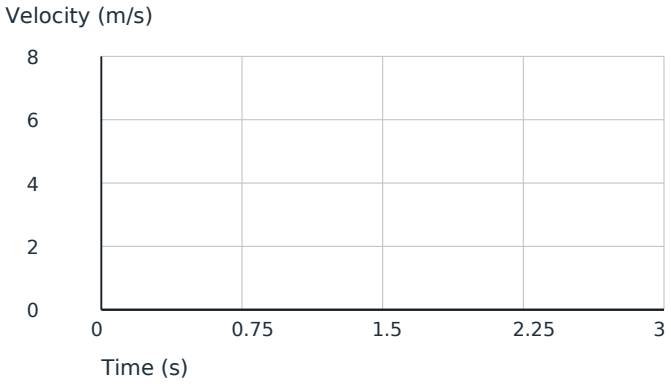
## 02 Worked example: explain the prediction

**Reference:** $F_{g,\ parallel} = mg \sin\theta$; $F_{g,\ perpendicular} = mg \cos\theta$
$\Sigma F = ma$.
Ramp angle θ above horizontal. Components replace the weight vector; they are not extra forces.

**Worked reasoning:** A **2 kg** cart initially at rest has a constant **4 N** net force right for **3 s**. Acceleration is **2 m/s²**; final speed is **6 m/s** right.

**Your turn:** Draw a force diagram and a velocity-time graph that agree. **Mark where mass enters the** reasoning.

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .




**Marzano check - circle:** ① ② ③

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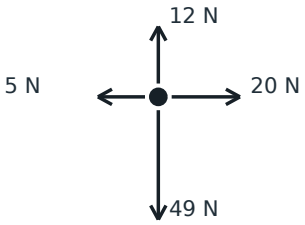
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Angled pull

A **5 kg** box stays on a level floor. A rope pulls with components **20 N** right and **12 N** up; friction is **5 N** left.
1. **Find normal force** and horizontal acceleration.
2. Use g=**9.8**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.



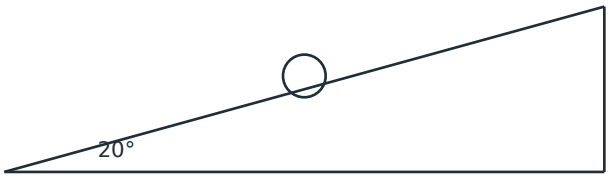

The dot represents the object. Normal not shown; find it

**Marzano check - circle:** 1 2 3

## 04 Frictionless ramp

A **2 kg** cart slides down a **20°** frictionless ramp.
1. **Find normal force** and acceleration along the ramp.
2. Use g=**9.8 m/s²**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.



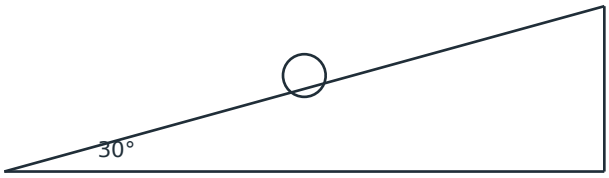

The marker represents the object on the sloping surface.

**Marzano check - circle:** 1 2 3

## 05 Oppose upward motion

A **4 kg** block slides up a **30°** ramp after release. Kinetic friction is **5 N** down the ramp. Use g = **9.8 m/s²** and **choose up the ramp as** positive. **Complete the force diagram**, calculate acceleration, and **describe the initial change** in speed.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




**Marzano check - circle:** 1 2 3

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Continue down each column.

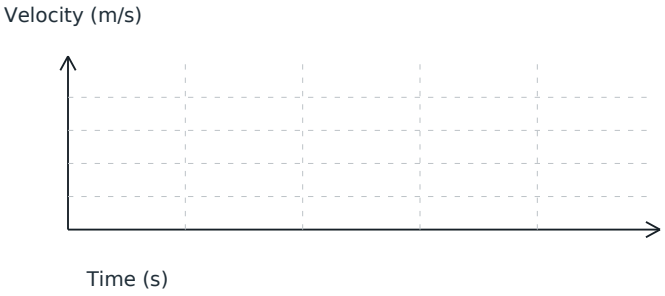
Sentence starters are optional. You may plan in any language.

## 06 Review: connect an earlier idea

Velocity is **+5 m/s** for **7 s**, then **-4 m/s** for **5 s**. **Find displacement** and distance. **Explain the reference information needed** for an angle of **35°**.
**Sketch the velocity**–time graph; **label the given values**.

**Word help:** displacement = change from initial position to final position, with direction

**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .

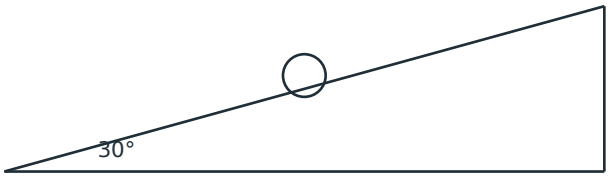



**Marzano check - circle:** 1 2 3

## 07 Exit check

A **2 kg** block slides down a **30°** ramp with **3.8 N** friction up the ramp.
1. **Find along-ramp acceleration** and normal force.
2. Use g=**9.8**.

**Sentence starter:** I found … [unit]. This value means … in this situation.



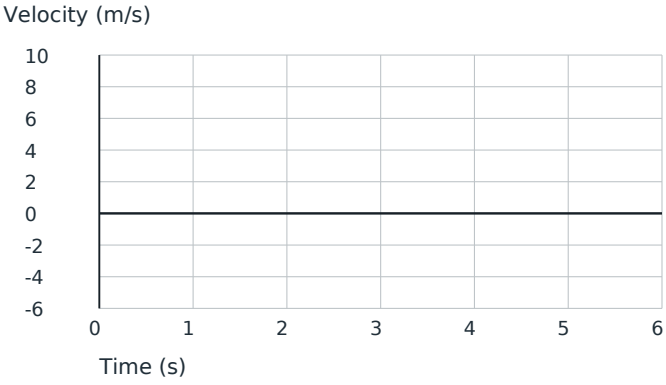

2 kg block slides down; friction 3.8 N up the ramp.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

A **2 kg** cart moving right at **8 m/s** has a constant **4 N** net force left for **6 s**. Find when it turns, final velocity, displacement and distance.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** 1 2 3